Capstone Defence — Quick Guide

Z-Empire Auto Accessories Shop Management and Pricing System · Sean Esperancilla · Read it the morning of the defence.

18	42	74	476	24	108	25
modules	tables in 6 schemas	products	stored prices	controllers	views	defects documented

THE THREE THINGS THAT MAKE THIS SYSTEM YOURS

Pricing has four keys	A price is Product × Tint Variant × Vehicle Class × Panel. Clear Series alone holds 134 stored prices. The **shade does not change the price** — it changes which roll the film is cut from.
Film is measured, not counted	A bolt is one unit in, one unit out. Film leaves the shelf in centimetres, and how many depends on which panel of which size of car. The cut-size table is the shop's own measurements.
Shade stock is derived, not stored	The ledger records a product and nothing finer, so no shade has a running balance. **Left = what the last count found** – **what has gone since.** It is shown as an estimate, with the ledger's own total beside it.

QUESTIONS YOU SHOULD EXPECT — AND THE HONEST ANSWER

Q **Why no database migrations?**

A The database existed first and holds the shop's live data. Generating it from code would mean regenerating what is already in use. Changes are made on the database and the model follows.

Q **Why is shade stock an estimate rather than a fact?**

A Because the stock ledger has no shade column, and adding one to a live database was not a change I was prepared to make mid-project. It is on my recommendations as the single change that would fix it.

Q **Is Waterfall really what you did?**

A No — **iterative waterfall**. 74% of source files were modified again in a later week, 49% in three or more weeks, and 25 defects each sent work back a phase. Royce's 1970 paper included feedback between adjacent phases; the one-pass version is a later simplification.

Q **How do you know staff cannot do an admin's job?**

A Every restriction was tested by typing the address directly, not by looking for a missing button. Deleting is Admin-only across all 46 delete endpoints, enforced by one convention so a new one is covered automatically.

Q **Is it actually deployed?**

A Yes — on live hosting with the shop's real catalogue restored. Four faults only appeared once it was somewhere real: hosting model, sign-in keys, protecting shop data on redeploy, and credentials off the repository.

Q **What would you do differently?**

A Name the panels better. “Full Wrap” means all the glass and “Whole Vehicle” means the painted body; both read as the whole car, and my own audit found prices entered against the wrong one.

BEFORE YOU WALK IN

- ☐ Open the live site once and sign in — do not let the first load of the day happen in front of the panel.
- ☐ **Generate one PDF receipt.** It is the riskiest untested piece: the PDF engine is a 64-bit binary and the app pool was set to 32-bit.
- ☐ Have a job order with a film line ready to show, and a vehicle checklist with a photo attached.
- ☐ Know your three numbers: 42 tables, 476 prices, 18 modules.
- ☐ If something breaks live, say what it is and move to the next screen. A fault you name is a bug; a fault you hide is a surprise.

If you are asked something you do not know: say so, say how you would find out, and move on. A panel forgives a gap far more readily than a guess that unravels.